

Before starting

1. The dataset

2. Descriptive Study of the Data

3. Linear regression

3.1 Simple linear regression

3.2 Multiple linear regression

4. Analysis of Variance Model

4.1. One factor analysis of variance model (Anova 1).

4.2. Two factor analysis of variance model (Anova 2).

4.3 Ancova

Course Outline

Linear Models and Their Generalizations

Code

Lecture 0: Introduction

Katia Meziani

Before starting



Before starting !

- Katia Meziani: meziani@ceremade.dauphine.fr (mailto:meziani@ceremade.dauphine.fr) (Office P207)
- To contact me: via Teams
- 5 practical sessions using R starting in November (with project submission)
- Lectures + Tutorials: outputs provided using R
- Evaluation
 - Midterm, Exam, Project (R)
 - Grade: max(0.4 Continuous Assessment + 0.6 Exam; E) with CC = 0.2 Project + 0.8 Midterm
- Recommendation letters if grade ≥ 17 : remember to manage/control your Facebook, LinkedIn pages,...
- Course materials + Tutorial sheets available online on the Teams group (not on Moodle)

Good luck!

1. The dataset

- We are focusing on the following dataset, sourced from the Economics Web Institute (<https://www.economicswebinstitute.org/data/wagesmicrodata.xls>).
- In this dataset, the hourly wage (in dollars) of 534 individuals was recorded along with other economic characteristics such as:
 - occupation (divided into 6 categories: 1 =Management, 2 =Sales, 3 =Clerical, 4 =Service, 5 =Professional, 6 =Other),
 - EDUCATION =number of years of education,
 - EXPERIENCE =number of years of experience,
 - AGE =age,
 - SEX =gender (1=female, 0=male),
 - MARR =marital status,
 - ...

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```
data1 <- read.csv("wagesmicrodata.csv", sep = ";",dec = ',',row.names = 1)
library(rmarkdown)
paged_table(data1)
```

	WAGE	OCCUPATION	SECTOR	UNION	EDUCATION	EXPERIENCE	A...	SEX	MA...	
	<dbl>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	
1	5.10	6	1	0	8	21	35	1	1	
2	4.95	6	1	0	9	42	57	1	1	
3	6.67	6	1	0	12	1	19	0	0	
4	4.00	6	0	0	12	4	22	0	0	
5	7.50	6	0	0	12	17	35	0	1	
6	13.07	6	0	1	13	9	28	0	0	
7	4.45	6	0	0	10	27	43	0	0	
8	19.47	6	0	0	12	9	27	0	0	
9	13.28	6	1	0	16	11	33	0	1	
10	8.75	6	0	0	12	9	27	0	0	

1-10 of 534 rows | 1-10 of 12 columns

Previous123456...54Next

Some variables have not been declared correctly; it is important to do so.

Hide

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4.3 Ancova

Course Outline

```
data1$OCCUPATION = as.factor(data1$OCCUPATION)
data1$SECTOR = as.factor(data1$SECTOR)
data1$UNION = as.factor(data1$UNION)
data1$MARR = as.factor(data1$MARR)
data1$SOUTH = as.factor(data1$SOUTH)
data1$RACE = as.factor(data1$RACE)
data1$SEX = as.factor(data1$SEX)
```

```
str(data1)
```

```
## 'data.frame':    534 obs. of  11 variables:
##  $ WAGE      : num  5.1 4.95 6.67 4 7.5 ...
##  $ OCCUPATION: Factor w/ 6 levels "1","2","3","4",...: 6 6 6 6 6 6 6 6 6 6 ...
##  $ SECTOR    : Factor w/ 3 levels "0","1","2": 2 2 2 1 1 1 1 1 2 1 ...
##  $ UNION     : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 1 1 ...
##  $ EDUCATION : int   8 9 12 12 12 13 10 12 16 12 ...
##  $ EXPERIENCE: int   21 42 1 4 17 9 27 9 11 9 ...
##  $ AGE       : int   35 57 19 22 35 28 43 27 33 27 ...
##  $ SEX       : Factor w/ 2 levels "0","1": 2 2 1 1 1 1 1 1 1 1 ...
##  $ MARR      : Factor w/ 2 levels "0","1": 2 2 1 1 2 1 1 1 2 1 ...
##  $ RACE      : Factor w/ 3 levels "1","2","3": 2 3 3 3 3 3 3 3 3 3 ...
##  $ SOUTH     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
```

Objectives of this Study:

- From the perspective of the economist, demographer, etc.
- Evaluate the potential effect of socio-demographic characteristics on employees’ wages.

Objectives of the Course:

- From the statistician's perspective.
- Study the statistical tools that allow for a rigorous analysis of the data (estimators, confidence intervals, statistical tests, etc.).

2. Descriptive Study of the Data

Before any elaborate statistical analysis, we must perform a descriptive study of the data. The quantities to compute and the graphical plots to check depend on the type of the variables (qualitative or quantitative).

The command `summary` provides the bellow quantitative summary. For qualitative variable, it provides a frequency table, while for quantitative variable, it provides the min, max, 1st quartile, median and 3rd quartile.

Hide

```
summary(data1$WAGE)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.     Max.
##  1.000    5.250    7.780    9.024   11.250   44.500
```

Hide

```
summary(data1$OCCUPATION)
```

```
##      1      2      3      4      5      6
##   55   38   97   83  105  156
```

Representation of Quantitative Variables: Histograms and Boxplots

- In a histogram, the rectangles have the class boundaries as their limits, and the height represents either the count or the frequency.
- Summarizes several characteristics of the position and dispersion of the variable studied (median, quartiles, minimum, maximum, or deciles).



Note!

- The boxplot is shortened to the minimum $x_{(1)}$ and the maximum $x_{(n)}$ when no observations fall beyond the *whiskers*.
- Conduct a careful examination of “outlier” individuals: aberrant values (entry errors, etc.) or extreme values (far from the center), **no systematic approach**.

Hide

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4. Analysis of Variance Model

4.1. One factor analysis of variance model (Anova 1).

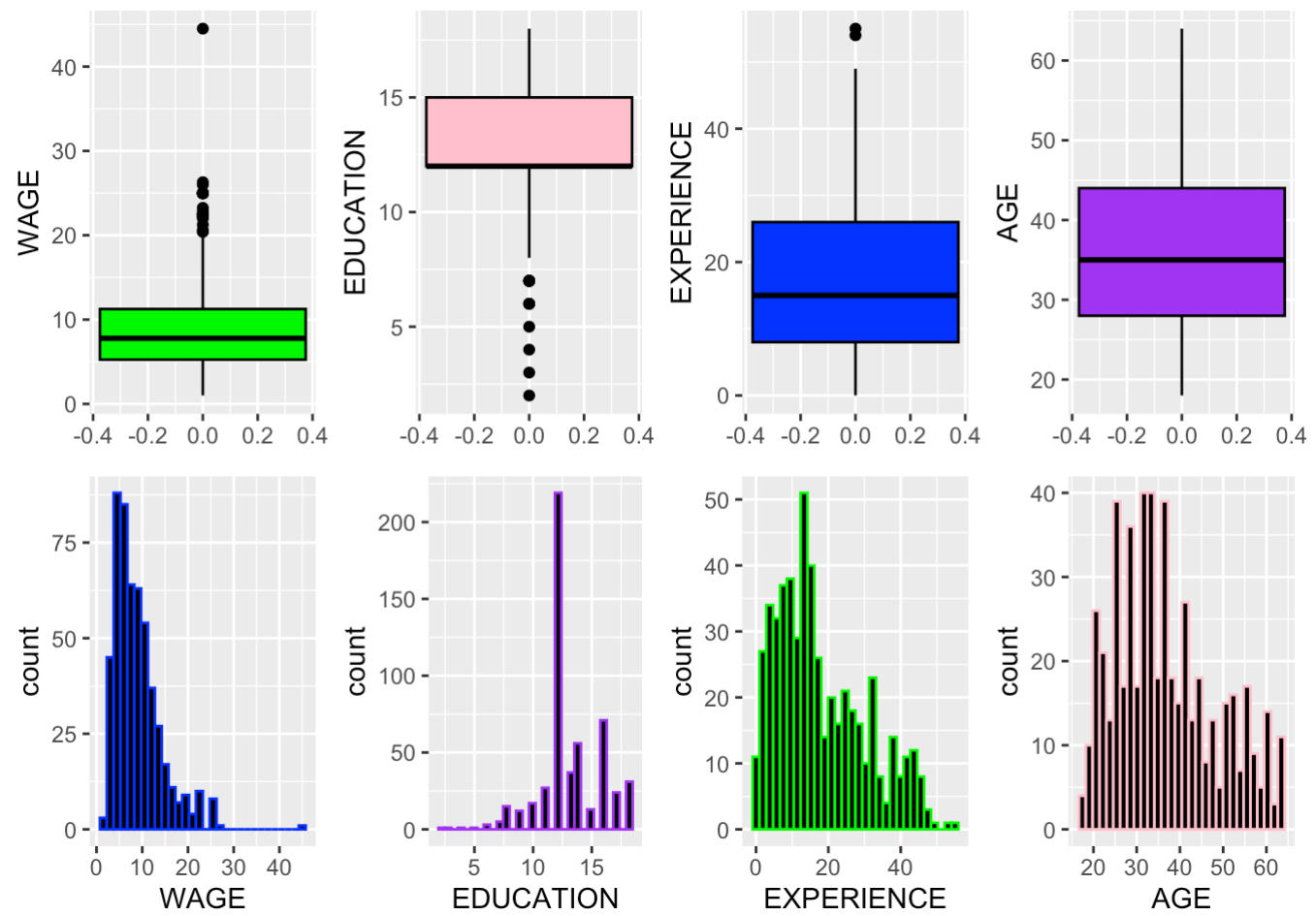
4.2. Two factor analysis of variance model (Anova 2).

4.3 Ancova

Course Outline

```
library(gridExtra)
library(ggplot2)
p <- ggplot(data1)
p1 <- p + geom_boxplot(aes(y=WAGE),col="black",fill="green")
p2 <- p + geom_boxplot(aes(y=EDUCATION),col="black",fill="pink")
p3 <- p + geom_boxplot(aes(y=EXPERIENCE),col="black",fill="blue")
p4 <- p + geom_boxplot(aes(y=AGE),col="black",fill="purple")

p5 <- p + geom_histogram(aes(x=WAGE),col="blue",fill="black")
p6 <- p + geom_histogram(aes(x=EDUCATION),col="purple",fill="black")
p7 <- p + geom_histogram(aes(x=EXPERIENCE),col="green",fill="black")
p8 <- p + geom_histogram(aes(x=AGE),col="pink",fill="black")
grid.arrange(p1, p2, p3,p4,p5,p6,p7,p8, ncol=4, nrow = 2)
```

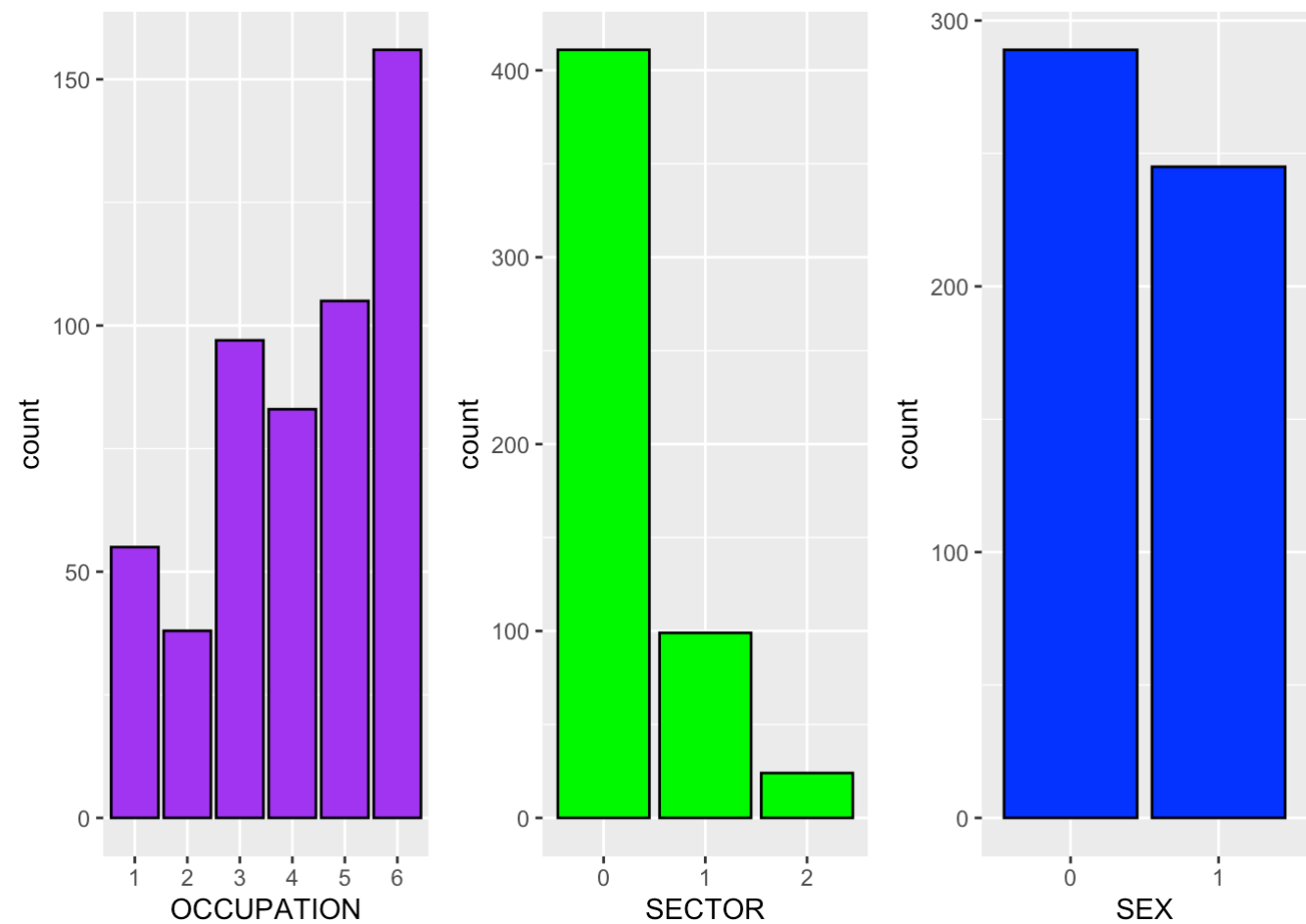


Representation of Qualitative Variables: Bar plot

The height of each bar is proportional to the count (or frequency). The width of each bar is arbitrarily chosen.

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```
p1 <- p + geom_bar(aes(x=OCCUPATION),col="black",fill="purple")
p2 <- p + geom_bar(aes(x=SECTOR),fill="green",col="black")
p3 <- p + geom_bar(aes(x=SEX),fill="blue",col="black")
grid.arrange(p1, p2, p3, ncol=3, nrow = 1)
```



3. Linear regression

We aim to understand the influence of other variables on salary (WAGE).

Before starting

- 1. The dataset
 - 2. Descriptive Study of the Data
 - 3. Linear regression
 - 3.1 Simple linear regression
 - 3.2 Multiple linear regression
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 - 4.3 Ancova
- Course Outline

First try to understand the influence of quantitative variables (EDUCATION , AGE , EXPERIENCE) on the variable to explain (WAGE).

Pearson correlation coefficient The linear relation between two quantitative variables X and Y can be quantified by Pearson’s correlation coefficient

$$\rho_{X,Y} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}, \quad r_{X,Y} = \frac{\frac{1}{n} \sum_{i,j} (y_i - \bar{y})(x_j - \bar{x})}{\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}}$$

where $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.

- By construction, $|\rho_{X,Y}| \leq 1$.
- If the points are perfectly aligned then $|\rho_{X,Y}| = 1$.

Let’s see what this looks like graphically.

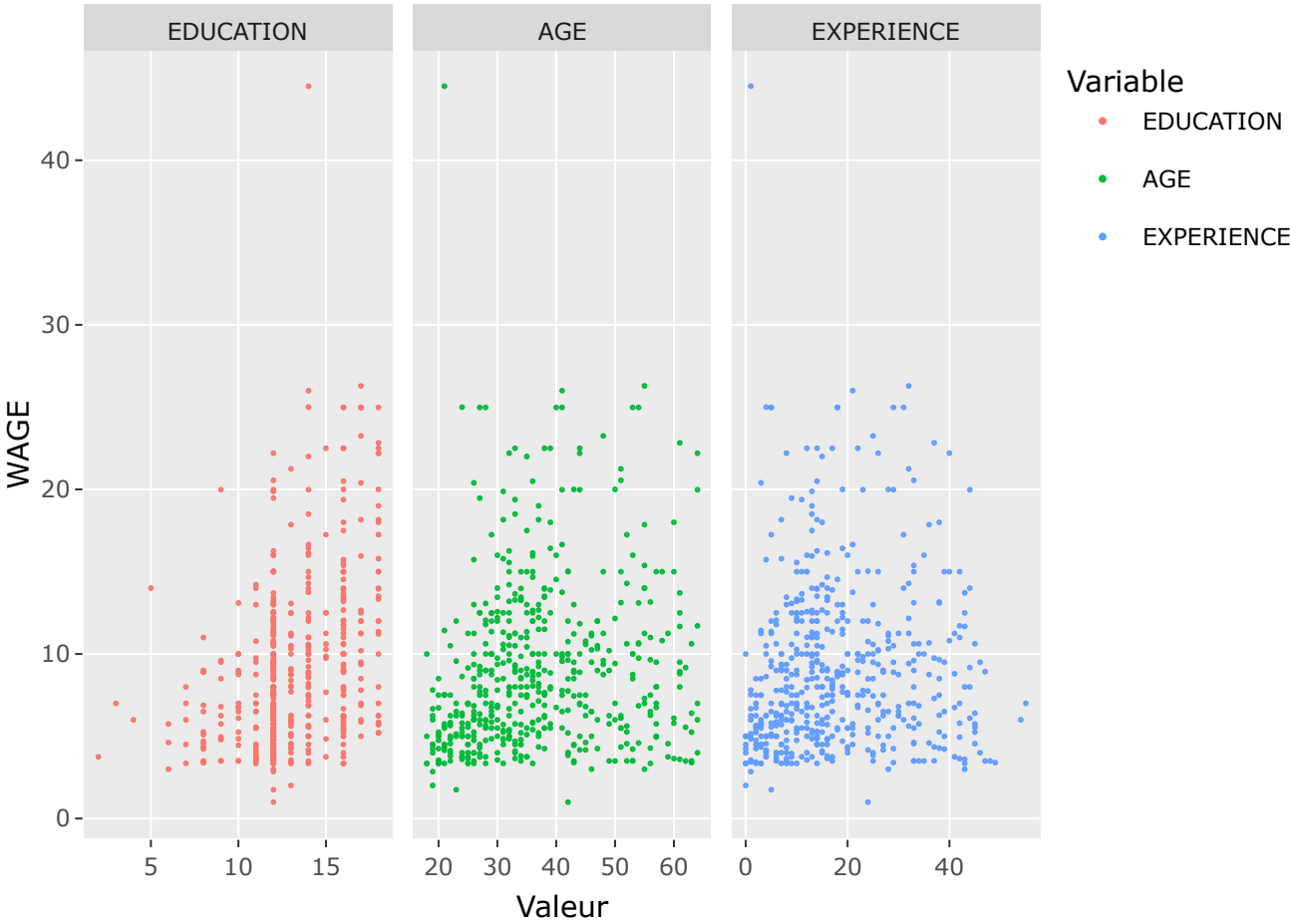
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```
library(ggplot2)
library(plotly)
library(reshape2)

# Conversion onto a dataframe
data_long <- melt(data1, id.vars = "WAGE",
                  measure.vars = c("EDUCATION", "AGE", "EXPERIENCE"),
                  variable.name = "Variable", value.name = "Valeur")

# Création du graphique
p_lm <- ggplot(data_long, aes(x = Valeur, y = WAGE)) +
  geom_point(aes(color = Variable), size = 0.25) +
  facet_wrap(~ Variable, scales = "free_x") +
  labs( y = "WAGE")

ggplotly(p_lm)
```



For these data, we get

Hide

```
cbind(cor(data1$WAGE,data1$EDUCATION),cor(data1$WAGE,data1$AGE),cor(data1$WAGE,data1$EXPERIENCE))
```

```
##           [,1]      [,2]      [,3]
## [1,]  0.3819221 0.1769669 0.08705953
```

Question: Is this big? small? significant?

We can make a test!

Tests:

H_0 : No link vs H_1 : link

- **Parametric tests (TP)** : powerfull by under conditions.

Before starting

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 - 3.2 Multiple linear regression
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 - 4.3 Ancova
- Course Outline

- Non-parametric tests (TNP) : less powerfull by the conditions are weaker.

 Good to know!

Small p -value $\Rightarrow$ reject of H_0 then link Lien
Large p -value $\Rightarrow$ non reject of H_0 then no link

Numerical X & Numerical Y

Spearman Test (corr lation). $H_0: corr_{X,Y} = 0$ vs $H_1: \overline{H_0}$.
Command **R**: `cor.test(X,Y,method="spearman")`

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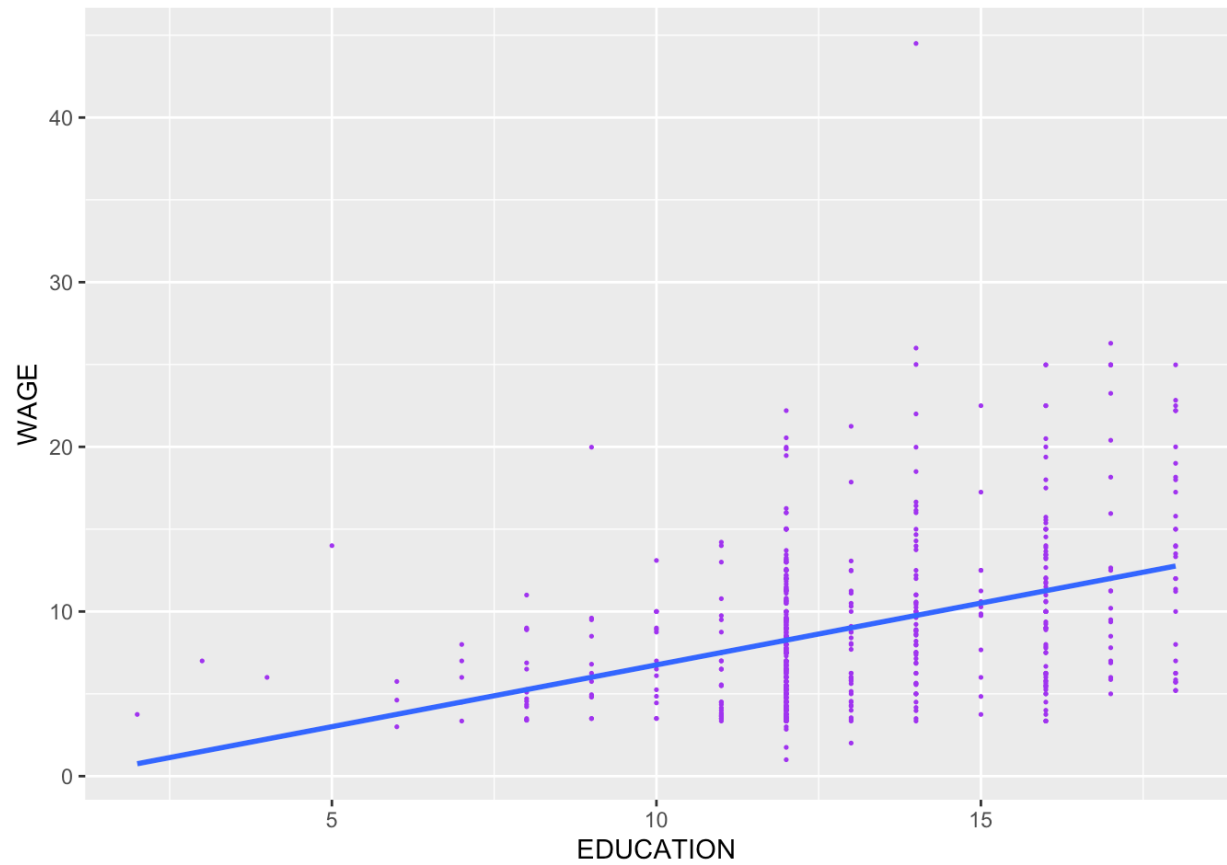
```
cbind(cor.test(data1$WAGE,data1$EDUCATION,method="spearman")$p.value,cor.test(data1$WAGE,data1$AGE,method="spearman")$p.value,cor.test(data1$WAGE,data1$EXPERIENC,method="spearman")$p.value)
```

```
##           [,1]           [,2]           [,3]  
## [1,] 6.291379e-20 3.361801e-09 8.71766e-05
```

 **Small p -value, we reject H_0 then there are linked.

3.1 Simple linear regression

We now return to the following scatterplot.



If we wish to summarize the scatterplot by a straight line, which will be called the simple linear regression line, we should write:

For the individual i ,

$$\text{Wage}_i = \beta_0 + \beta_1 \text{Education}_i + e_i$$

where e_i is an error term between the line and observation y_i .

 In this very simple model, we are explain the (quantitative) variable of interest by a quantitative *explanatory variable*, also called a *covariate*.

- The slope (β_1) and the intercept (β_0) of the line are estimated from the observations to correctly "locate" the line.
- In this course, we shall study how to estimate these parameters, and the properties of the estimators.
- We shall also study how to decide whether the slope is significantly different from 0, i.e. we shall write statistical tests on the model parameters.

3.2 Multiple linear regression

We can also try to explain the wages as a linear combination of the other quantitative covariates:

Before starting

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- 3. Linear regression
 - 3.1 Simple linear regression
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 - 4.3 Ancova

Course Outline

$$\text{Wage}_i = f(\text{Education}_i, \text{Age}_i, \text{Experience}_i) + e_i$$
$$\text{Wage}_i = \beta_0 + \beta_1 \text{Education}_i + \beta_2 \text{Age}_i + \beta_3 \text{Experience}_i + e_i$$

$$Y_i = \beta_0 + \sum_{k=1}^K \beta_k x_i^k + e_i$$

where x_i^k s the value of the k -th explanatory variable for individual i .

This is now multiple linear regression (see Chapters 1 and 2). The same questions arise: significance of the β_k , selection of most relevant covariates, etc.

4. Analysis of Variance Model

4.1. One factor analysis of variance model (Anova 1).

- Study the relationship between WAGE (quantitative variable of interest) and qualitative variables, such as SEX or OCCUPATION (type of job).
- Graphically, boxplots can be created for each category of the qualitative variable.

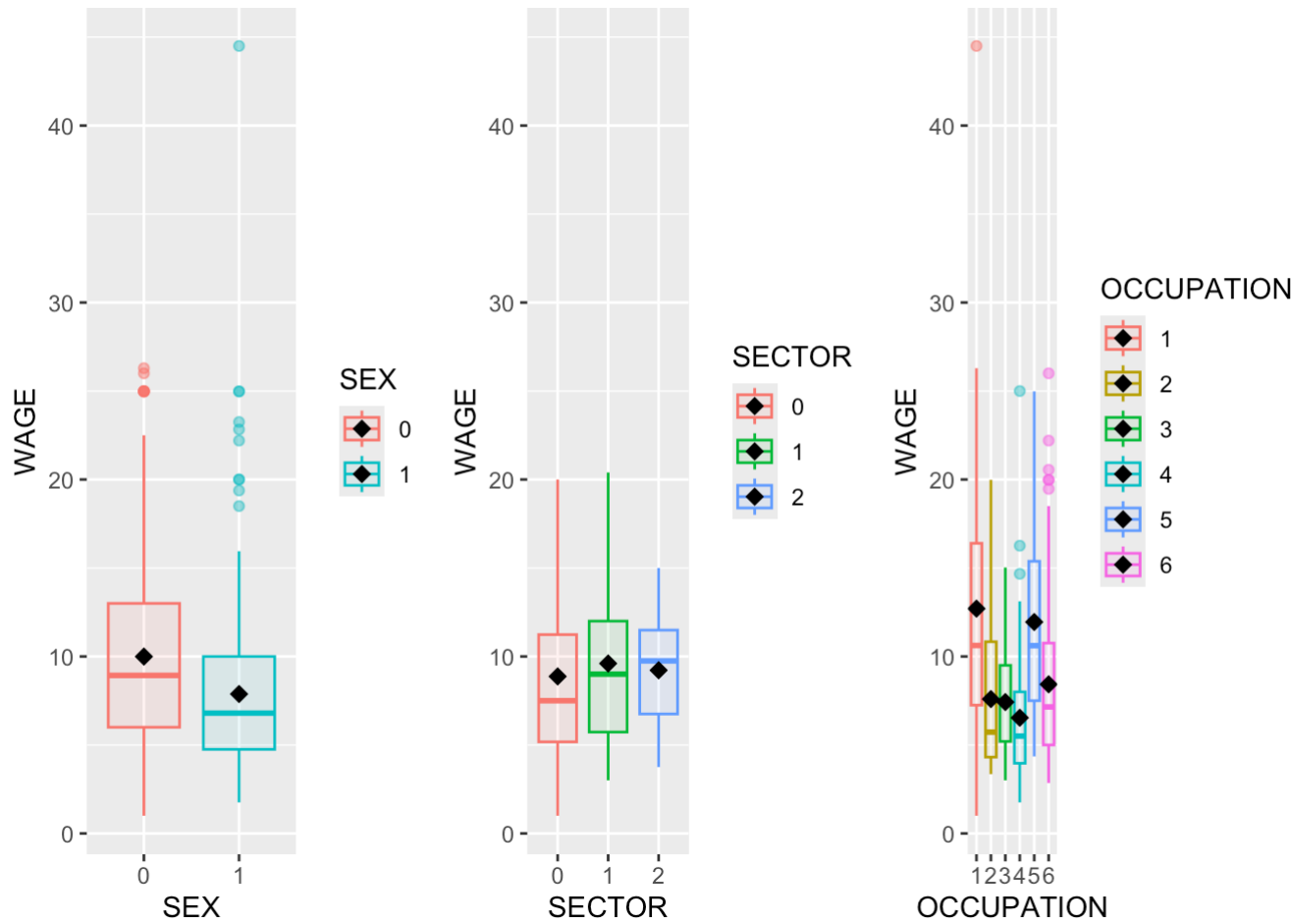
Hide

```
a1= ggplot(data1, aes(y=WAGE, x=SEX,colour=SEX ,fill=SEX))+ geom_boxplot(alpha=0.1,outlier.alpha=0.5)+ stat_summary(fun=mean, colour="black", geom="point",shape=18, size=3)

a2= ggplot(data1, aes(y=WAGE, x=SECTOR,colour=SECTOR ,fill=SECTOR))+ geom_boxplot(alpha=0.1, outlier.alpha=0)+ stat_summary(fun=mean, colour="black", geom="point",shape=18, size=3)

a3= ggplot(data1, aes(y=WAGE, x=OCCUPATION,colour=OCCUPATION ,fill=OCCUPATION))+ geom_boxplot(alpha=0.1, outlier.alpha=0.5)+ stat_summary(fun=mean, colour="black", geom="point",shape=18, size=3)
```

```
grid.arrange(a1, a2, a3, ncol = 3, nrow = 1)
```



It is indeed possible to write a model to study WAGE based on SEX or OCCUPATION': $\text{Wage}_i = \underbrace{\beta_0}_{\text{Men's wage}} + \beta_1 \text{Sex}_i + \beta_2 \text{Occupation}_i + e_i$

In this course, this model is presented in the form of a linear model (Chapter 3 of the course).

4.2. Two factor analysis of variance model (Anova 2).

We might wonder whether there is a joint effect from both factors. We will then examine at the wages as follows:

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```
tapply(data1$WAGE, list(data1$SEX, data1$OCCUPATION), mean)
```

##	1	2	3	4	5	6
## 0	13.72176	9.495714	7.489048	7.226471	12.77396	9.068175
## 1	11.05619	5.241765	7.404211	6.059388	11.10500	5.731333

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 - 3.2 Multiple linear regression
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 - 4.1. One factor analysis of variance model (Anova 1).
 - 4.2. Two factor analysis of variance model (Anova 2).
 - 4.3 Ancova
- Course Outline

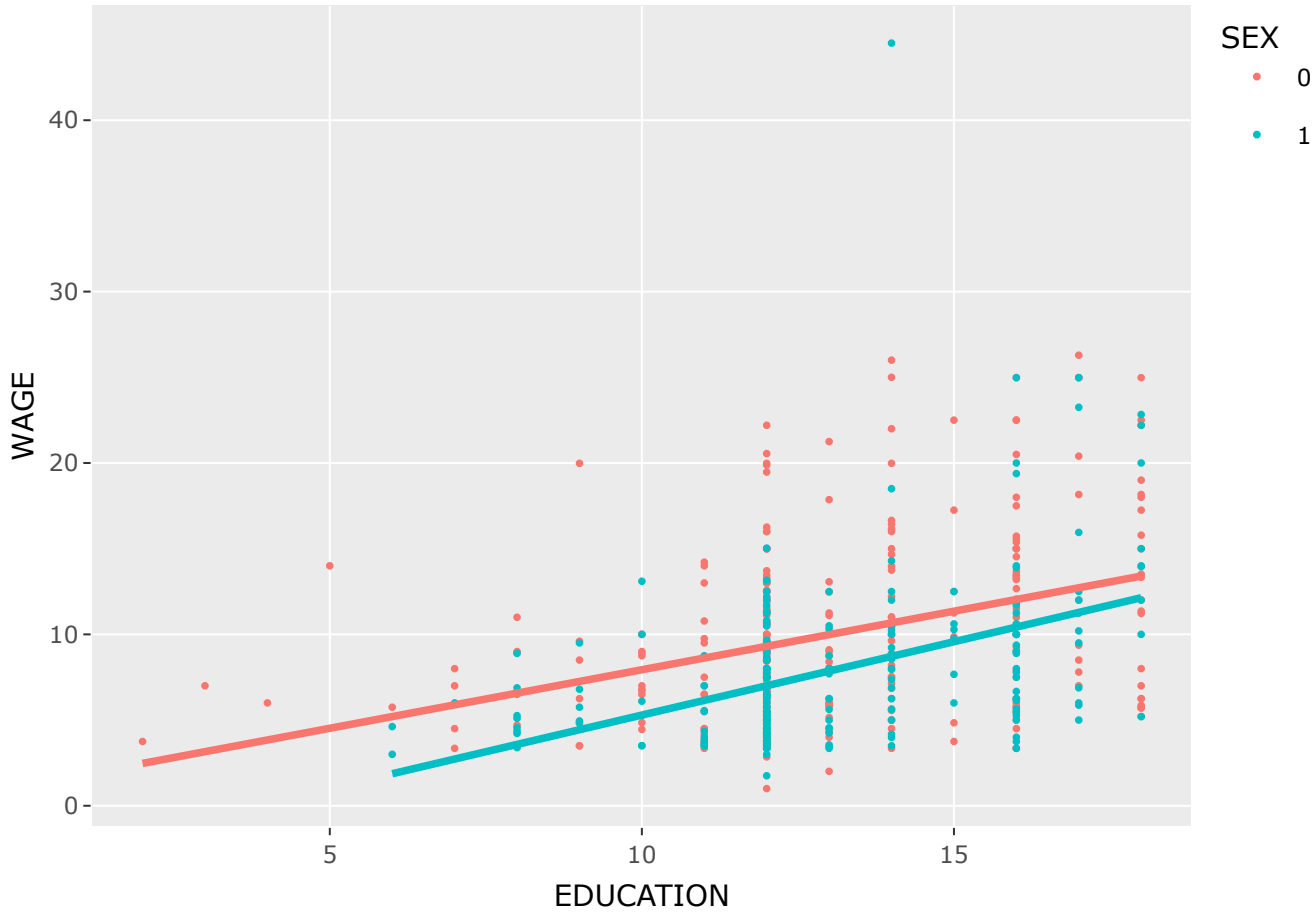
We will study the influence of each factor on WAGE , but also there joint influence (potential interactions). We shall write a linear model to this end in chapter 3.

4.3 Ancova

We might think that the relationship between WAGE and EDUCATION differs depending on whether one is a man or a woman.

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```
plot1<-ggplot(data1) + aes(x = EDUCATION, y = WAGE, color = SEX) + geom_point(size = 0.5)+ geom_smooth( method = "lm",se = FALSE)
ggplotly(plot1)
```



Similarly, we shall see in Chapter 3 that we can write a linear model corresponding to this question.

Course Outline

- 1. Linear Model. Mathematical Properties, Assumptions, Tests
 - Very mathematical
 - Based on algebra (orthogonal projectors)
 - Will be reused in all subsequent chapters
- 2. Specific Issues in Regression. Tests, Variable Selection, Examples in R
 - Application.
 - Handling examples
 - Interpretation of R outputs
- 3. ANOVA with One and Two Factors, ANCOVA
- 4. Generalized Linear Models
 - Addresses cases with non-continuous response variables (e.g., 0 — 1, counting...)
 - Very practical
- 5. Bayesian Inference (Dauphinois Specificity)